

ECE-501 Week-2 Group-11

1. What is YieldSAT?

The YieldSAT is a multimodal benchmark dataset for high resolution agricultural yield analysis. It integrates crop yield data derived from agricultural machinery with data from satellite and environment. The dataset comprises Sentinel-2 optical, multispectral imagery, as well as yield and other parameters including weather, soil and topographic data. The yield measurements have been processed at a spatial level of 10 m * 10 m, and aligned with the Sentinel-2 images. YieldSAT was also useful for studying agricultural fields based on information provided at the field and sub-field level, as obtained by remote sensing and machine learning techniques. For our project, YieldSAT will provide the information for the on-the-field agriculture analysis and optical data for the crop classification task.

2. What data does it provide?

YieldSAT offers a number of agri- and remote-sensing-related data types. Here the primary data: multispectral optical data from Sentinel-2, yield measurements, weather data, soil data and topographic data. Optical data from the Sentinel-2 satellites offers information on the agricultural fields via various optical bands that can be used to identify features pertinent to crop type and vegetation. Yield measurements are taken with combine harvesters fitted with yield-monitoring and GPS systems and are aggregated onto a 10 m spatial grid. Weather data includes information on the environment (e.g., temperature, precipitation) and soil data and topographic data include additional information about the growing environment. In our project the information from the Sentinel-2 data source, as well as the crop related information, will be of great relevance, whereas the SAR–optical feature fusion experiment will be supported by Sentinel-1 SAR data which will be acquired separately.

3. What countries are available?

Agricultural field data for four countries Argentina, Brazil, Uruguay and Germany are included in YieldSAT. The data include four major crop types: wheat, soybean, rapeseed and corn. We chose Group B (Argentina and Brazil) for our project. Corn, soybean and wheat are the relevant crop types in these two

countries. Thus, in this project we will especially concentrate on these two countries and take available agricultural fields and crop data of these two countries as a basis for the classification experiment. The choice of Argentina and Brazil allows multiple crop classes to be used to evaluate and compare optical-only, SAR-only and fused SAR–optical classification.

4. What are crop labels/yield masks used for?

An Agricultural field contains a type of crop identified by a label, such as corn, soybeans or wheat. Crop classification is a supervised machine learning task and the label (ground truth) is the signature used for comparison to the model's output, which is a class prediction. Yield information are spatial yield measurements from fields associated with the agricultural fields. Because yield is measured in a spatially referenced grid on top of Sentinel-2 imagery, the produced yield mask can be used to identify the pixels where yield measurements are available and exclude the pixels where no measurements for yield are available. This facilitates the choice and interpretation of valid farming areas and the related ground truth data. In our project, crop labels will be the main class label tools, and by attaching masks and yield information, understanding and identifying important field areas will be better understood.

5. What is Sentinel-1 ?

- Sentinel-1 is a Copernicus Earth-observation satellite mission operated by the European Space Agency (ESA). It carries a C-band Synthetic Aperture Radar (SAR) instrument. The important point for our project is that Sentinel-1 provides radar imagery during both day and night and in all weather conditions. Unlike optical satellites, it does not depend on sunlight and is much less affected by clouds. Sentinel-1 uses a frequency of about 5.405 GHz (C-band). The current Sentinel-1 mission includes Sentinel-1A, 1C and 1D; Sentinel-1C was launched in December 2024 and Sentinel-1D in November 2025. We are interested in Sentinel-1 because its radar observations can provide information about vegetation and crop fields, which we can combine with optical information for crop classification.

6. What is SAR?

- SAR= Synthetic Aperture Radar.

SAR is an active remote sensing system. Instead of waiting for sunlight and measuring reflected sunlight like an optical satellite, SAR:

- Sends microwave energy towards Earth's surface.
- The energy interacts with the ground/vegetation.
- Some of the energy is scattered back.
- The satellite measures this returned signal called backscatter.

As SAR provides its own source of illumination, it can operate during both daytime and nighttime. Its microwave signals are also much less affected by clouds than optical wavelengths. Therefore, SAR is useful for obtaining consistent observations of agricultural areas throughout the crop growing season.

7. What does VV mean?

- VV stands for Vertical transmit and Vertical receive. The radar signal is transmitted using vertical polarization and the returning signal is also received using vertical polarization. Therefore, it is a co-polarization.

8. What does VH mean?

- VH stands for Vertical transmit and Horizontal receive. The radar signal is transmitted using vertical polarization, while the returning signal is received using horizontal polarization. Therefore, it is a cross - polarization.

9. Why is SAR useful for crop classification?

- The physical structure, the plant height, the leaf arrangement, the biomass and the moisture characteristics are different for different crops. The properties are those which influence the interaction of radar signals with vegetation and with the earth's surface. This means that the radar backscatter values can vary between different crops. Another change in radar response occurs as a crop matures. So, multiple Sentinel-1 observations can deliver a temporal development signature or pattern, which can be useful for crop identification. SAR is used particularly in crop classification because,
 - It can take observations both during the day and at night.
 - Depends much less on clouds as an obstruction to optical imagery.
 - It gives information on the vegetation structure and moisture.
 - Sentinel-1 can be used to monitor time series of crop growth.
 - VV and VH are orthogonal polarization components which can be used as classification features.

10. How will GEE be used?

- Google Earth Engine (GEE) offers access to vast archives of satellite and geospatial data, without having to download and process all the satellite images ourselves.

In GEE, for our project, we can access Sentinel-1 GRD directly via `ee.ImageCollection('COPERNICUS/S1_GRD')`

We can then:

- Select the study area.
- Choose the desired time frame.
- Process the Sentinel-1 images with filtering based on acquisition parameters.
- Choose images with VV and VH polarization.
- Get VV and VH features.
- Add these capabilities to those of Sentinel-2.
- Apply the fused features to classify the crops.

The following is a basic example of filtering using Sentinel-1.

```
var s1 = ee.ImageCollection('COPERNICUS/S1_GRD')
  .filterDate('2025-06-01', '2025-09-30')
  .filter(ee.Filter.eq('instrumentMode', 'IW'))
  .filter(ee.Filter.listContains(
    'transmitterReceiverPolarisation', 'VV'))
  .filter(ee.Filter.listContains(
    'transmitterReceiverPolarisation', 'VH'))
  .select(['VV', 'VH']);
print(s1);
```

The code imports the Sentinel-1 GRD collection, filters it for the desired date range, requires both VV and VH polarisations and picks out the VV and VH bands.

11. Project Overview and Methodology

Our project focuses on classifying crops using Sentinel-1 SAR and Sentinel-2 optical satellite data from the YieldSAT dataset. We selected Argentina and Brazil as our study areas and will focus on three crops: corn, soybean, and wheat. The main goal is to understand how well different types of satellite data can distinguish between these crop types.

We will compare three approaches: optical-only, SAR-only, and SAR + optical fusion. Sentinel-2 data will provide spectral and vegetation information, while Sentinel-1 will provide information related to crop structure and moisture. We

plan to use features such as vegetation indices, VV, VH, VV/VH, and texture features. The data will be processed using Google Earth Engine, and the YieldSAT crop labels will be used as the ground truth for classification.



Preliminary Feature Set

Data Source

Planned Features

Sentinel-2 Optical

NDVI, NDRE, NDMI, EVI, GLCM texture features

Sentinel-1 SAR

VV, VH, VV/VH ratio, GLCM texture features

- These handcrafted features will be extracted and used to construct optical-only, SAR-only, and fused feature sets for subsequent machine learning experiments.

12. Work Done This Week

During Week 2, the project methodology and implementation plan were further refined. The YieldSAT dataset and its relevance to crop classification were studied, along with Sentinel-1 SAR data and its VV and VH polarizations. The use of Google Earth Engine for acquiring Sentinel-1 data was also investigated. Relevant literature on Sentinel-1, Sentinel-2, SAR-optical fusion, and crop classification was reviewed. Based on this study, the preliminary optical and SAR feature sets and the overall experimental workflow were finalized.

13. Work Planned for Next Week

The next phase will focus on moving from the planned methodology to initial implementation. The group will organize the required YieldSAT data for Argentina and Brazil, identify the relevant crop labels and field information, and acquire corresponding Sentinel-1 SAR data using Google Earth Engine. Initial preprocessing and alignment of Sentinel-1 and Sentinel-2 data will then be performed, followed by implementation of the first handcrafted optical and SAR features.

14. Challenges Faced

The main challenges during Week 2 were understanding the structure and available information in the YieldSAT dataset, becoming familiar with Google Earth Engine and Sentinel-1 data acquisition, and planning the alignment of optical and SAR data for the same agricultural fields.

15. Literature Review

- a. Orynbaikyzy et al. (2020): This study looked at crop classification using Sentinel-1 and Sentinel-2 data and compared different combinations of the two sensors. It showed that the choice of features and the availability of optical data can affect classification performance. This paper is useful for our project because we are also comparing optical-only, SAR-only, and SAR + optical approaches.

- b. Tufail et al. (2021): The authors used Sentinel-1 and Sentinel-2 time-series data with a Random Forest classifier for crop type mapping. They compared SAR-only, optical-only, and combined data to see how each contributed to crop classification. This supports our plan to compare the three data configurations and also gives us a practical machine-learning baseline.

- c. Eisfelder et al. (2024): This study used Sentinel-1 and Sentinel-2 time-series data together with Google Earth Engine for crop classification and agricultural monitoring. It shows how satellite data from different sources can be processed over large areas using GEE. This is relevant to our project because we also plan to use GEE for Sentinel-1 data processing.

- d. Chen et al. (2020): This paper studied crop discrimination using Sentinel-1 SAR information along with texture features. It considered VV and VH polarization and used GLCM-based texture information to improve crop separation. The study is relevant to our project because we also plan to use VV, VH, and texture features as part of our SAR feature set.

- e. Liu et al. (2024): This study explored the integration of optical and SAR data for crop classification using a deep-learning-based fusion approach. The work highlights how combining information from different satellite sensors can provide more useful information for distinguishing crops. It supports our decision to investigate SAR and optical data fusion, while our project will determine its actual benefit through our own experiments.

16. Key Findings and Recommendations

Key Findings

The research shows that Sentinel-2 is useful for spectral and vegetation information, while Sentinel-1 provides information about crop structure and moisture. VV, VH, and texture features can also help in distinguishing different crop types. Overall, the literature suggests that combining SAR and optical data may provide more useful information than using only one data source.

Recommendations

We will compare optical-only, SAR-only, and SAR + optical approaches using the same crop classes and evaluation method. Multiple observations should be used where possible, and missing or invalid data should be handled carefully. The final comparison should use measures such as accuracy, precision, recall, F1-score, and a confusion matrix to see which approach performs better in our actual experiments.

17. Conclusion

Overall, the Week 2 research helped us understand how Sentinel-1 and Sentinel-2 data can be useful for identifying different crops. From the papers we studied, we found that optical data gives information about vegetation, while SAR data gives additional information about crop structure and moisture. This tells us why using both types of data could be useful. Based on our research, we will compare optical-only, SAR-only, and combined SAR + optical data for corn, soybean, and wheat in Argentina and Brazil. The next step will be to implement these approaches and see how well they perform.

References:

Orynbaikyzy, A., Gessner, U., & Conrad, C. (2020). *Crop Type Classification Using Fusion of Sentinel-1 and Sentinel-2 Data: Assessing the Impact of Feature Selection, Optical Data Availability, and Parcel Sizes on the Accuracies*. Remote Sensing, 12(17), 2779.

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